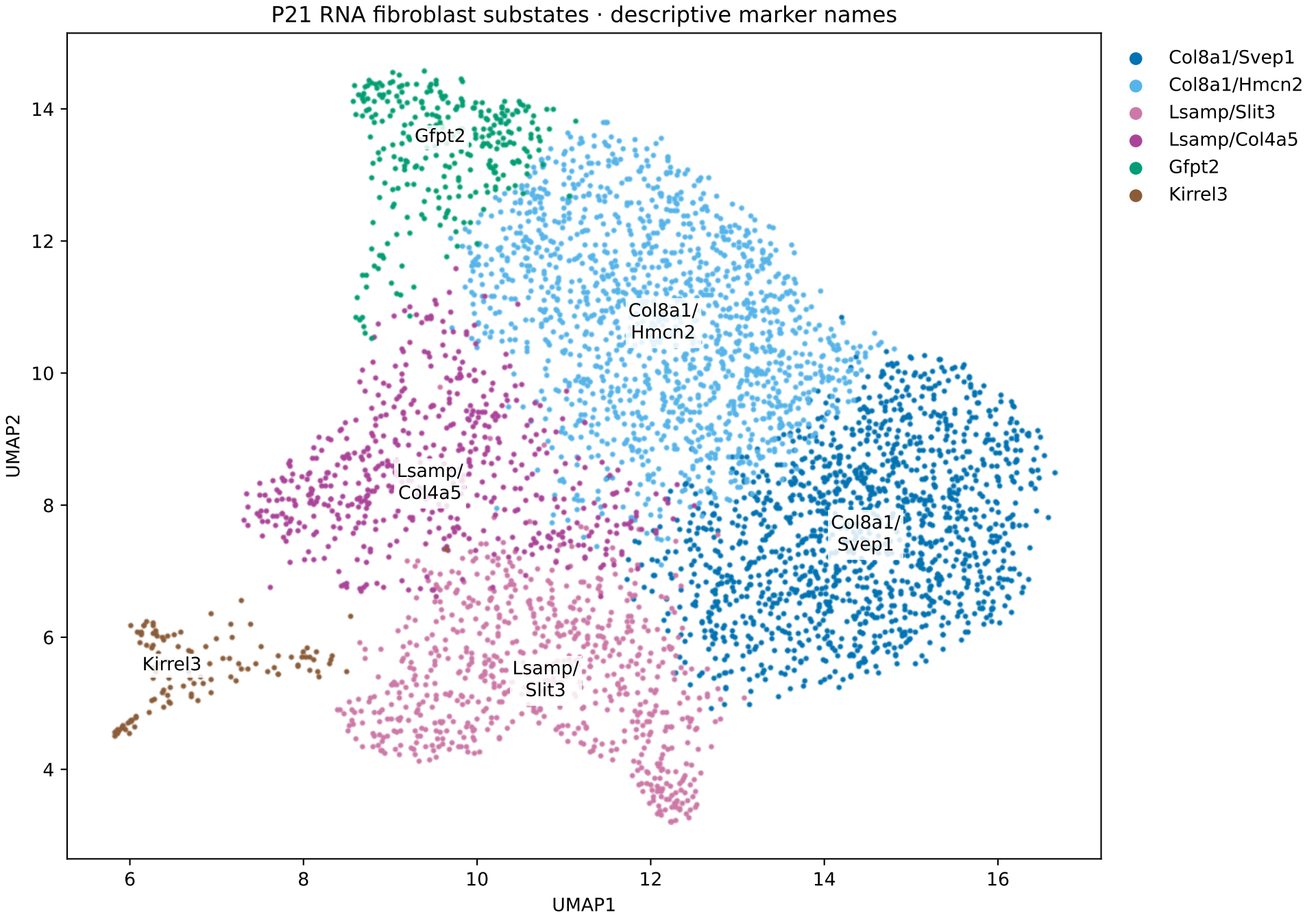
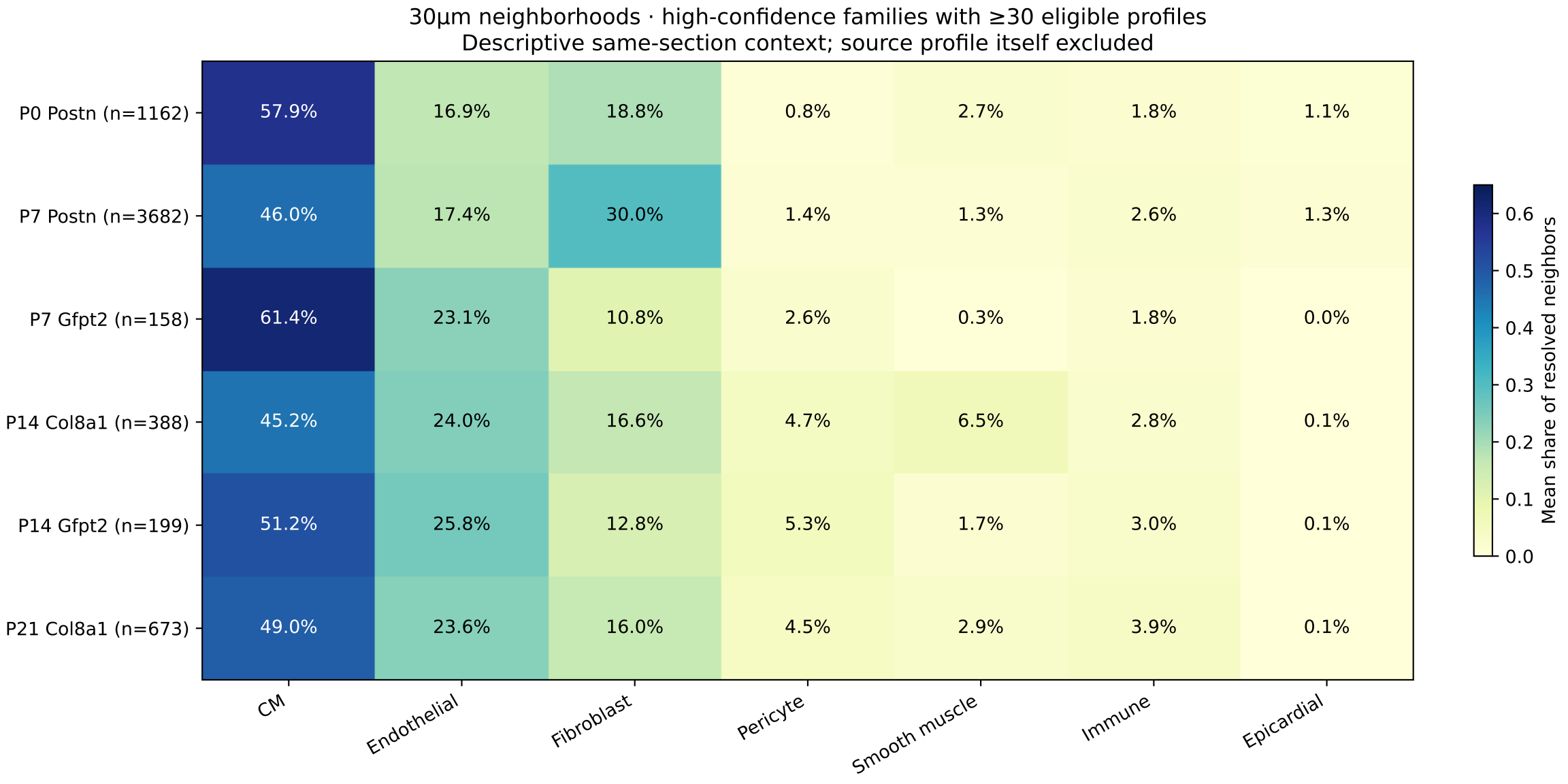
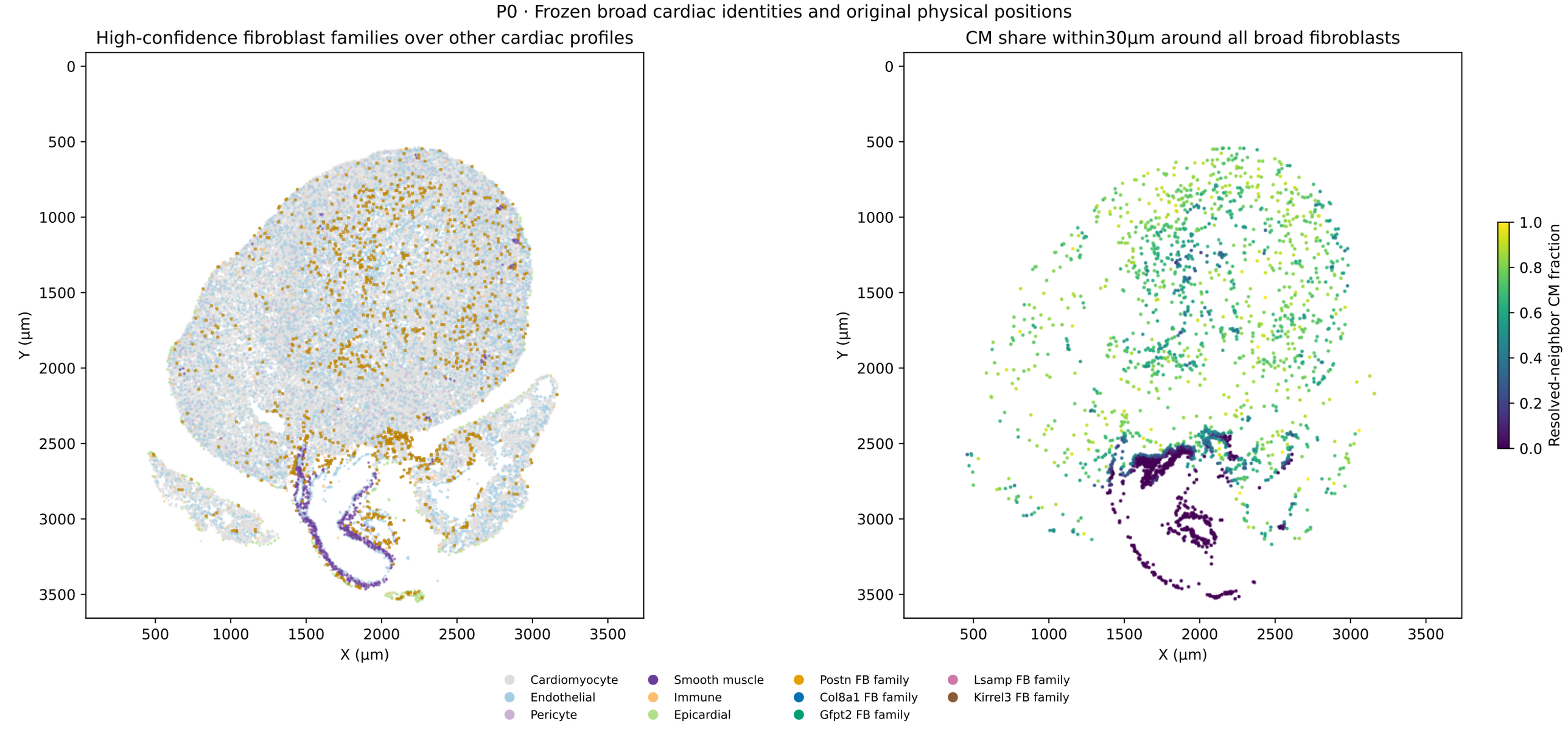
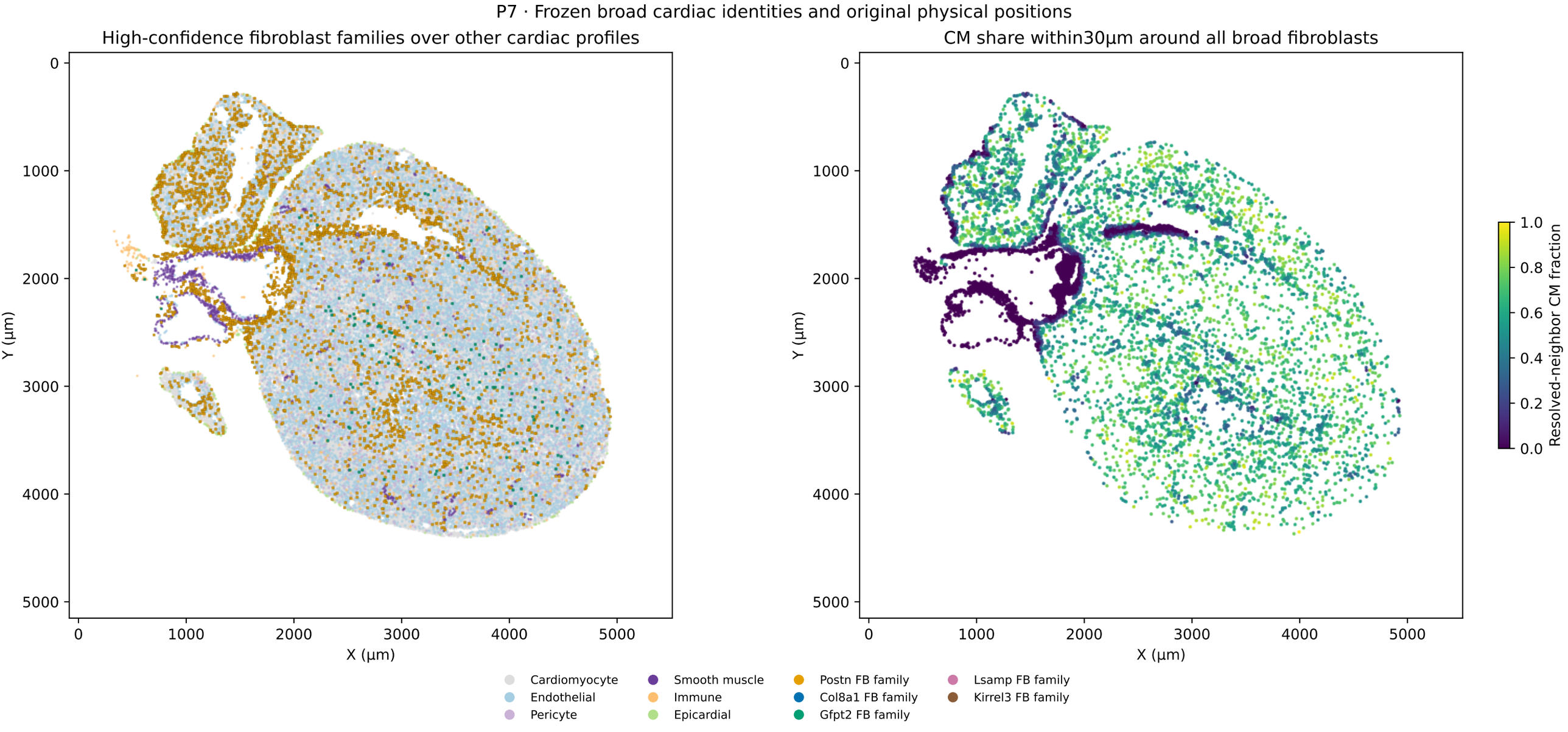
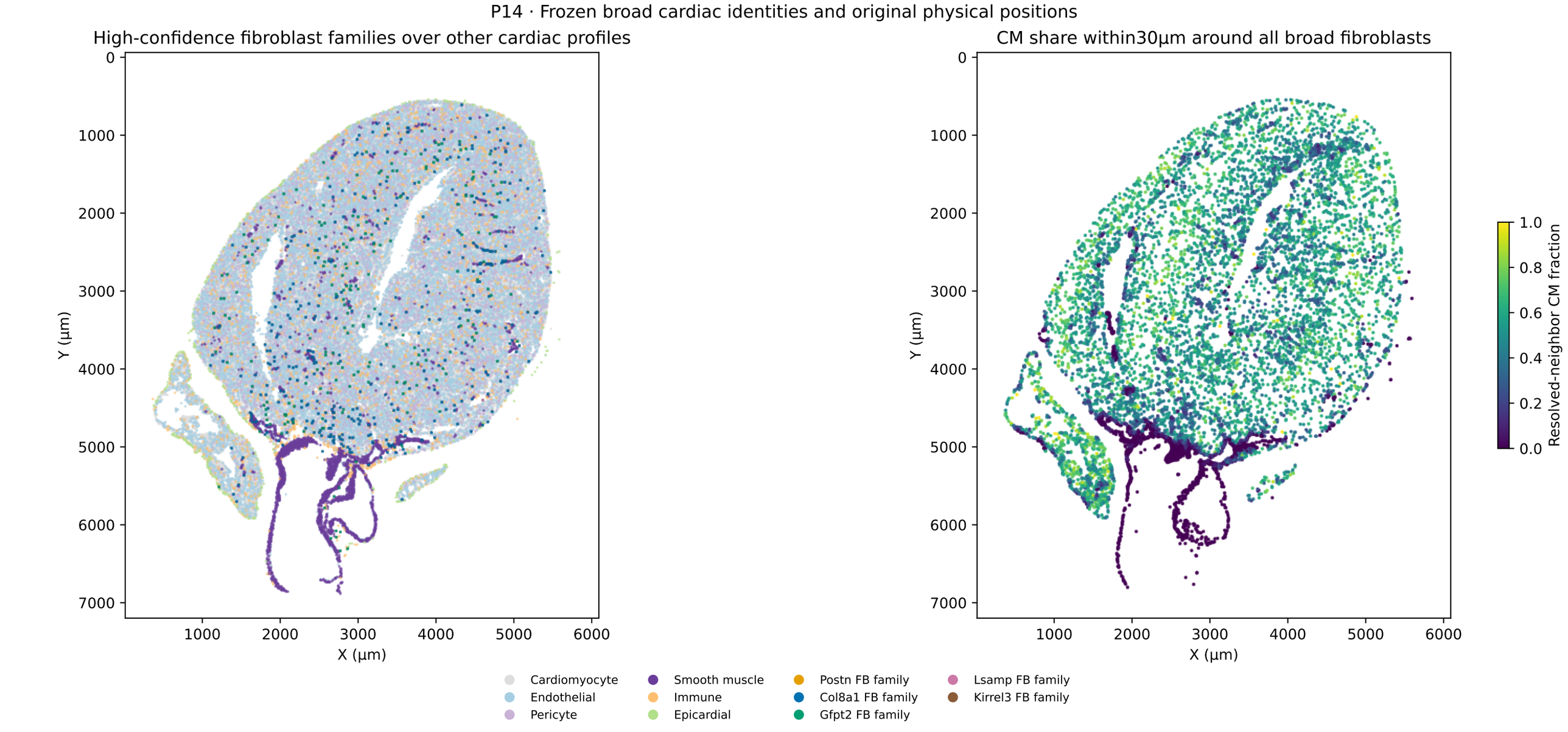
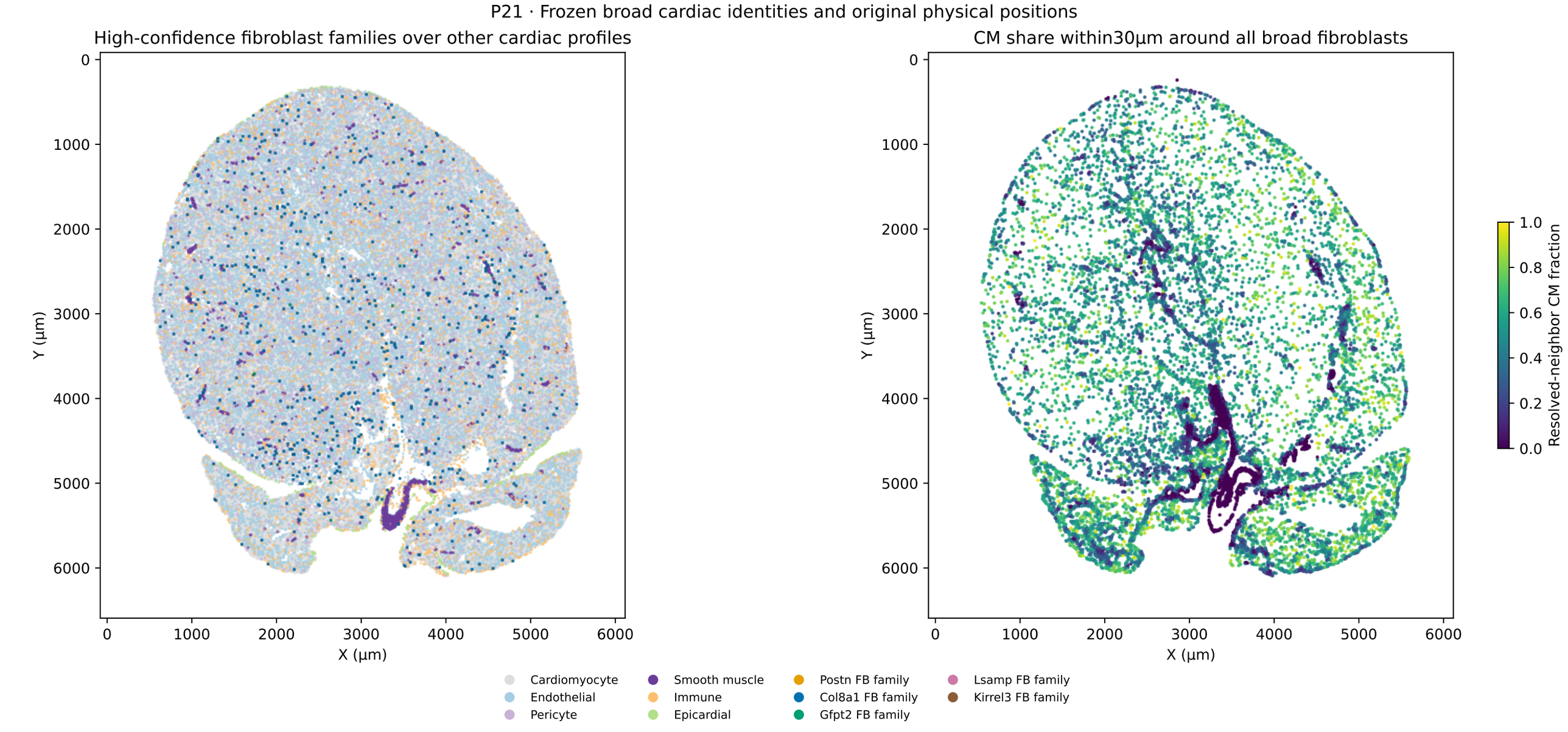

P21 RNA fibroblast substates · descriptive marker names
Col8a1/Svep1
Col8a1/Hmcn2
Lsamp/Slit3
Lsamp/Col4a5
Gfpt2
Kirrel3
Gfpt2
Col8a1/
Hmcn2
Lsamp/
Col4a5
Col8a1/
Svep1
Kirrel3
Lsamp/
Slit3
UMAP2
UMAP1
14
12
10
8
6
4
6
8
10
12
14
16


30µm neighborhoods · high-confidence families with ≥30 eligible profiles
Descriptive same-section context; source profile itself excluded
P0 Postn (n=1162)
P7 Postn (n=3682)
P7 Gfpt2 (n=158)
P14 Col8a1 (n=388)
P14 Gfpt2 (n=199)
P21 Col8a1 (n=673)
57.9%
16.9%
18.8%
0.8%
2.7%
1.8%
1.1%
46.0%
17.4%
30.0%
1.4%
1.3%
2.6%
1.3%
61.4%
23.1%
10.8%
2.6%
0.3%
1.8%
0.0%
45.2%
24.0%
16.6%
4.7%
6.5%
2.8%
0.1%
51.2%
25.8%
12.8%
5.3%
1.7%
3.0%
0.1%
49.0%
23.6%
16.0%
4.5%
2.9%
3.9%
0.1%
CM
Endothelial
Fibroblast
Pericyte
Smooth muscle
Immune
Epicardial
Mean share of resolved neighbors
0.0
0.1
0.2
0.3
0.4
0.5
0.6


P0 · Frozen broad cardiac identities and original physical positions
High-confidence fibroblast families over other cardiac profiles
CM share within30µm around all broad fibroblasts
Y (µm)
X (µm)
0
500
1000
1500
2000
2500
3000
3500
Resolved-neighbor CM fraction
0.0
0.2
0.4
0.6
0.8
1.0
Cardiomyocyte
Endothelial
Pericyte
Smooth muscle
Immune
Epicardial
Postn FB family
Col8a1 FB family
Gfpt2 FB family
Lsamp FB family
Kirrel3 FB family


P7 · Frozen broad cardiac identities and original physical positions
High-confidence fibroblast families over other cardiac profiles
CM share within30µm around all broad fibroblasts
Y (µm)
X (µm)
0
1000
2000
3000
4000
5000
Resolved-neighbor CM fraction
0.0
0.2
0.4
0.6
0.8
1.0
Cardiomyocyte
Endothelial
Pericyte
Smooth muscle
Immune
Epicardial
Postn FB family
Col8a1 FB family
Gfpt2 FB family
Lsamp FB family
Kirrel3 FB family


P14 · Frozen broad cardiac identities and original physical positions
High-confidence fibroblast families over other cardiac profiles
CM share within30µm around all broad fibroblasts
Y (µm)
X (µm)
0
1000
2000
3000
4000
5000
6000
7000
Resolved-neighbor CM fraction
0.0
0.2
0.4
0.6
0.8
1.0
Cardiomyocyte
Endothelial
Pericyte
Smooth muscle
Immune
Epicardial
Postn FB family
Col8a1 FB family
Gfpt2 FB family
Lsamp FB family
Kirrel3 FB family


P21 · Frozen broad cardiac identities and original physical positions
High-confidence fibroblast families over other cardiac profiles
CM share within30µm around all broad fibroblasts
Y (µm)
X (µm)
0
1000
2000
3000
4000
5000
6000
Resolved-neighbor CM fraction
0.0
0.2
0.4
0.6
0.8
1.0
Cardiomyocyte
Endothelial
Pericyte
Smooth muscle
Immune
Epicardial
Postn FB family
Col8a1 FB family
Gfpt2 FB family
Lsamp FB family
Kirrel3 FB family